

ENV 6106

Air Pollution Modeling

Spring 2023

CO Categorical Finding

- A way to determine air quality conformity of a transportation project
 - Categorical finding → screening → refined modeling
 - If pass, no need to do further screening modeling
 - See link below for more information
 - https://www.fhwa.dot.gov/environment/air_quality/conformity/policy_and_guidance/cmcf_2017/
- MOVES + CAL3QHC modeling → max CO conc
 - Technical details:
 - https://www.fhwa.dot.gov/environment/air_quality/conformity/policy_and_guidance/cmcf_2017/technical_document.pdf
 - Can use table or website:
 - https://www.fhwa.dot.gov/environment/air_quality/conformity/policy_and_guidance/cmcf_2017/tool.cfm

CO Categorical Finding

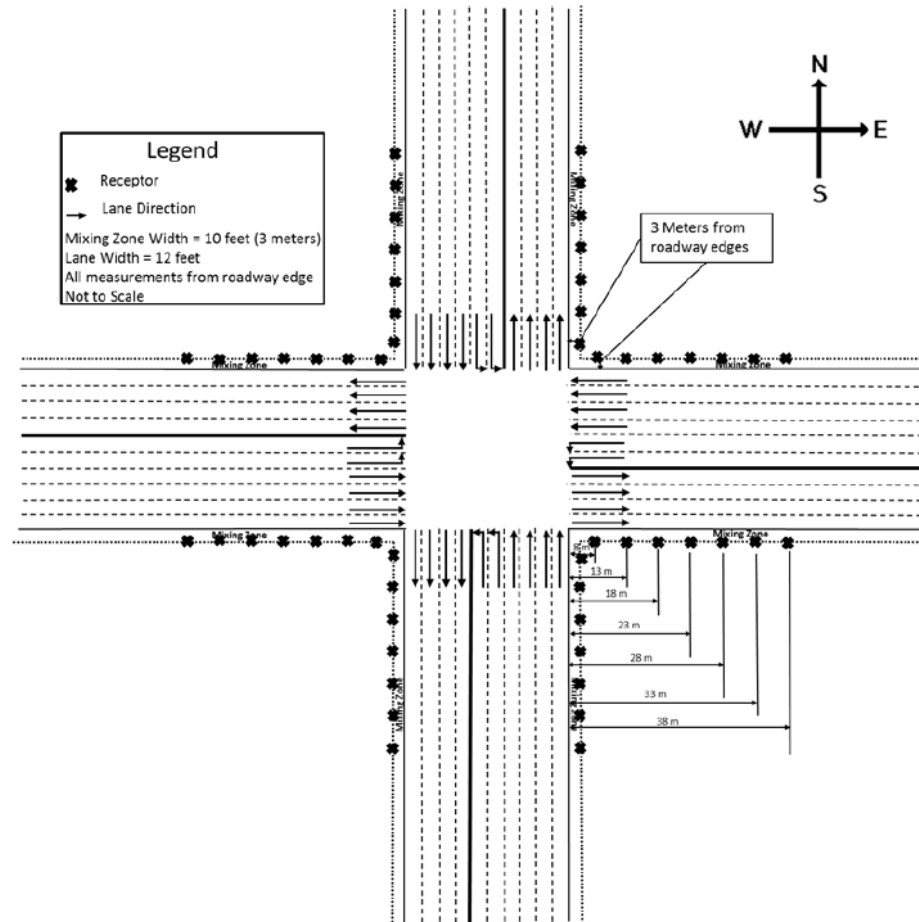


Figure 5. Additional Receptor Locations

CO Categorical Finding

Urban Intersection

Parameter	Acceptable Range
Analysis year	Greater than or equal to 2017
Angle of cross streets for intersection (degrees)	90
Maximum grade for the intersection (%)	Less than or equal to 2
Maximum grade on cross street for the intersection (%)	0
Number of through lanes	Less than or equal to 4
Number of left turn lanes	Less than or equal to 2
Lane width (ft)	12
Median width (ft)	0
Peak hour average approach speed (mph)	Greater than or equal to 25
Peak hour approach volume (vph)	Less than or equal to 2640
Peak hour Level of Service	A through E
Ambient temperature (°F)	Greater than or equal to -10
Heavy-duty diesel trucks (%)	Greater than or equal to 5
1-hour background CO concentrations (ppm)	Less than or equal to 32.6
8-hour background CO concentrations (ppm)	Less than or equal to 7.3
Persistence factor	Less than or equal to 0.7

Some Jargons

- Level of Service
 - How congested is the roadway
 - A (free flow); B (good flow); C (stable flow); D (almost unstable flow); E (at capacity); F (road = parking lot)
 - Described more in the “Highway Capacity Manual”
- Background CO concentration
 - CO not due to project, but still need to consider
- Persistence factor
 - A factor to convert peak 1-hour conc to peak 8-hour
- Ps: Diesel cars usually emit less CO than gas cars

Screening model

- CO Florida model is the screening model FDOT uses for intersections
 - A few other states also have their own model
- Very easy to use, user friendly.
- Developed by Dr. C. David Cooper's team at UCF
- Download here:
 - <http://www.fdot.gov/environment/software/software.shtm>
- Uses MOVES2010a + CAL3QHC